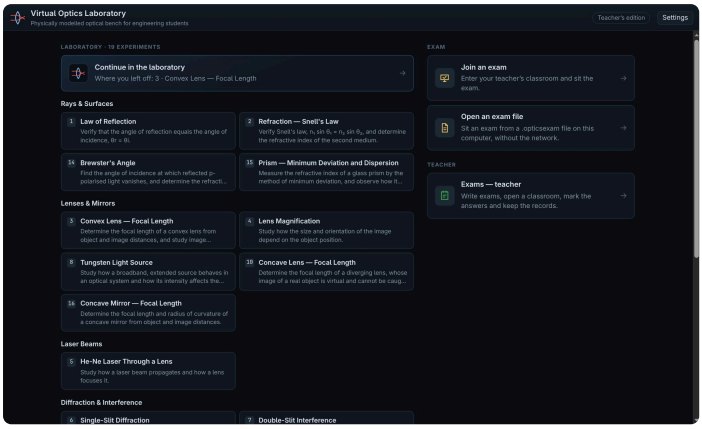
Virtual Optics Laboratory — quick guide

How to open an experiment, work on the bench, measure, and record results · 19 experiments

1. Starting

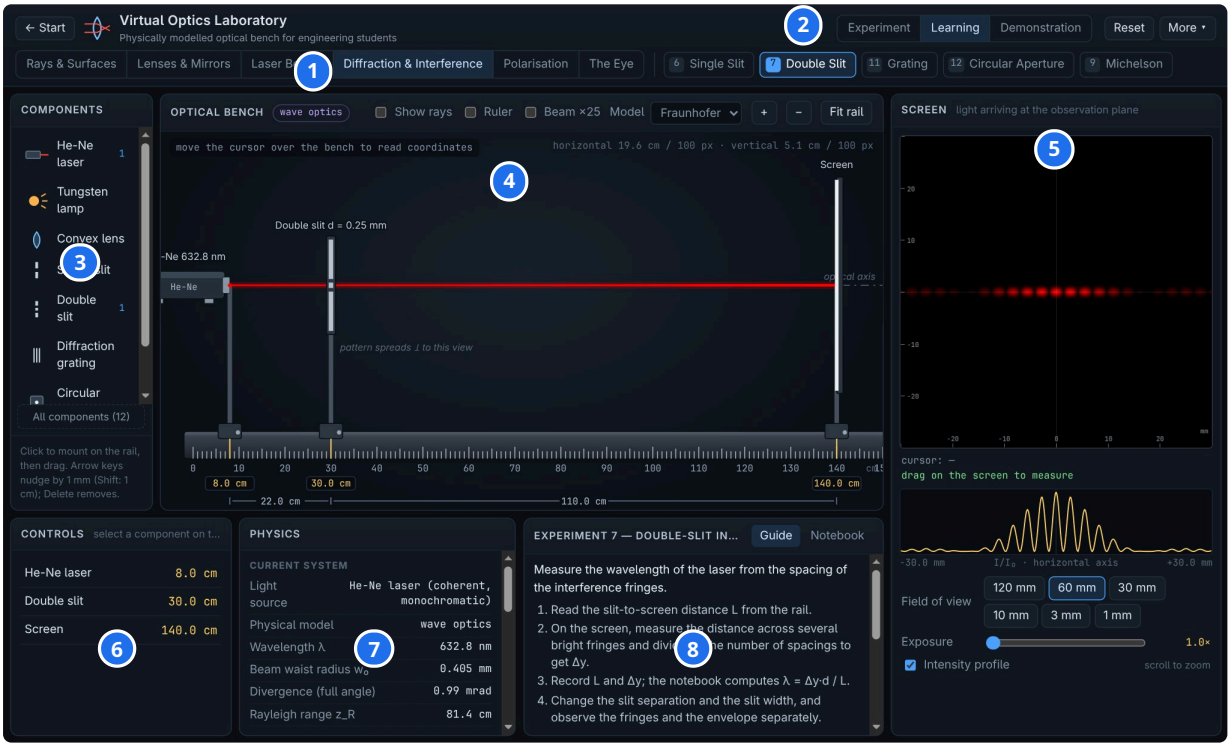


The start screen lists every experiment, grouped by topic. **Click a card** to open the laboratory on that experiment, or **Continue in the laboratory** to go back to where you left off: the laboratory remembers your last set-up.

On the right are the exam functions (section 5) and, at the top, **Settings**.

Inside the laboratory, ← **Start** brings you back here.

2. The workspace



The workspace with experiment 7, the double slit. Experiments on the optical bench all look like this; the others replace the bench by their own apparatus.

1 **Topics and experiments.** Pick a topic on the left; its experiments appear beside it.

2 **Mode, Reset, More.** *Reset* restores the starting layout of the experiment. *More* holds “Clear the bench”, “Save to a file…” and “Load from a file…”.

3 **Components.** Click one to mount it on the rail. The list shows the parts of the current topic; “All components” shows the rest.

4 **Bench.** Drag a component along the rail. Arrow keys move it by 1 mm (with Shift: 1 cm); Delete removes it. Positions and distances are printed under the rail.

5 **Screen.** The light that arrives at the screen. Drag on it to measure a length. “Field of view” zooms, “Exposure” brightens faint patterns.

6 **Controls.** Click a component on the bench to change its settings here, for example the focal length of a lens or the width of a slit.

7 **Physics.** The values of the current set-up: readings, and the calculated results.

8 **Guide and Notebook.** The steps of the experiment with its equations, and the table for your measurements.

3. Working on an experiment

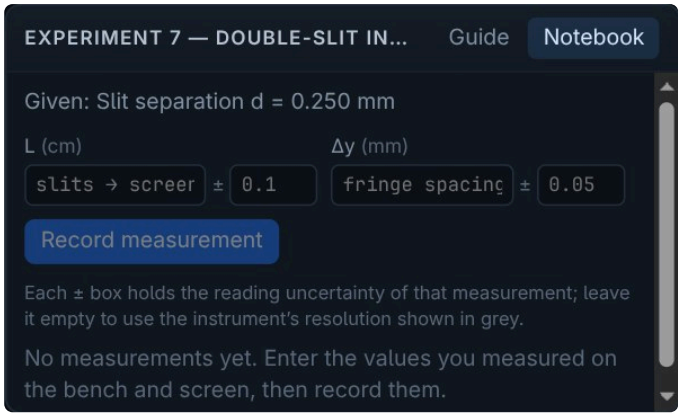
The three modes

- **Learning** shows every value, equation and assumption.
- **Experiment** withholds the calculated results (“to be measured”), so that you have to measure them. *Reveal theory* in the Physics panel shows them afterwards.
- **Demonstration** shows everything and adds a *Sweep* button that moves one setting slowly back and forth.

Measuring

- **Along the rail:** read the positions and distances under the rail, or tick *Ruler* and drag on the bench.
- **On the screen:** drag from one point to another.
- **Angles:** in the reflection and refraction experiments, read them from the protractor; “Align protractor to normal” sets its zero.
- **+**, **−** and *Fit rail* zoom the bench. Zooming never changes the physics.

The notebook



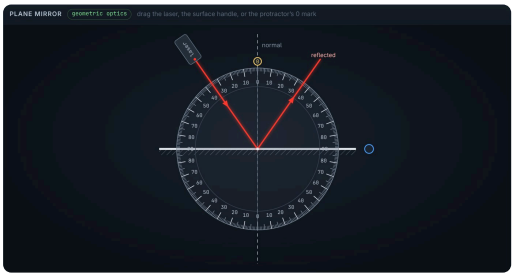
1. Open the **Notebook** tab and type your readings. The small ± box holds the uncertainty of a reading; leave it empty to use the value shown in grey.
2. Press **Record measurement**. The row shows the result calculated from your readings, with its uncertainty.
3. Repeat. In Experiment mode, **Check against theory** then compares your results with the accepted values; in the other modes the comparison is shown straight away.
4. **Export report** and **Export CSV** save your work; type your name beside them for the report.

If the laboratory shows a warning instead of a result, the set-up is outside what its models cover. It tells you so and does not draw a guess.

4. The experiments

Each picture shows the experiment as it opens. “Record” lists what you enter in the notebook; “Result” is what the notebook calculates from it.

RAYS & SURFACES



1 Law of Reflection

Drag the ray box around the mirror, or tilt the mirror, to set an angle of incidence. Align the protractor with the normal and read both angles from it.

Record: θi and θr, for at least five angles.
Result: θr = θi.



2 Refraction — Snell’s Law

Choose the two media and drag the ray box. Read the incident and the refracted angle from the protractor, both from the normal. Send the light from the denser medium to see total internal reflection.

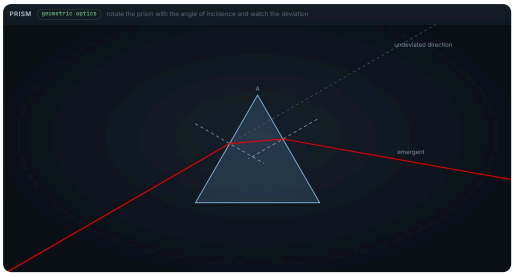
Record: θ_1 and θ_2.
Result: n_2 from $n_1 \sin\theta_1 = n_2 \sin\theta_2$.



14 Brewster’s Angle

Set the polarisation of the ray to p. Increase the angle of incidence slowly and watch the reflected power fall; find the angle at which the reflected ray disappears.

Record: θ_B.
Result: $n_2 = n_1 \tan\theta_B$.

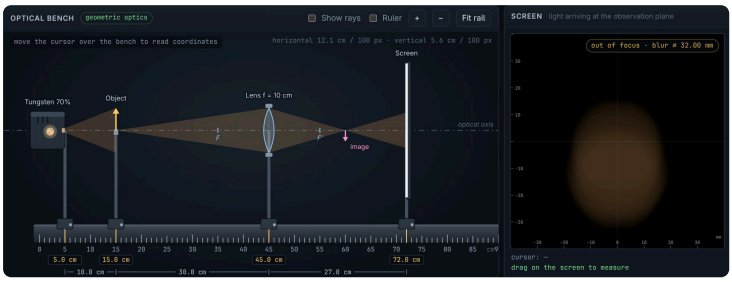


15 Prism — Minimum Deviation

Choose a laser line. Change the angle of incidence and watch the deviation reading δ; find the angle at which δ is smallest. Switch to white light to see the spectrum.

Record: δ_min.
Result: n = sin((A + δ_min)/2) / sin(A/2).

LENSES & MIRRORS



3 Convex Lens — Focal Length

Slide the screen along the rail until the image is sharpest. Read the object–lens and lens–screen distances from the rail. In Experiment mode, try “Use unknown lens”.

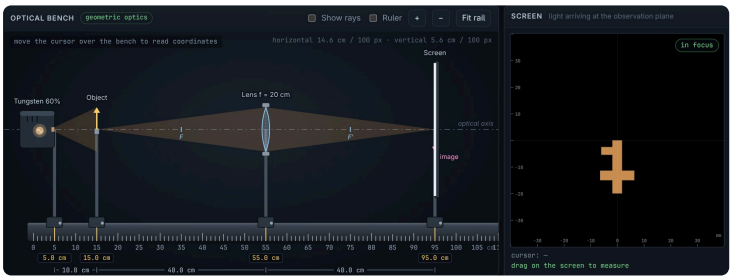
Record: do and di, for several object positions.
Result: 1/f = 1/do + 1/di.



4 Lens Magnification

Focus the image, then drag across it on the screen to measure its height (negative if it is inverted). Repeat with the object at other positions.

Record: ho, hi, do and di.
Result: m = hi/ho compared with −di/do.



8 Tungsten Light Source

Focus the image and change the lamp intensity: the brightness changes, the position and size of the image do not. Remove the object to image the lamp aperture itself.

Record: lamp %, do and di.
Result: 1/f = 1/do + 1/di.



10 Concave Lens — Focal Length

Drag the concave lens to the far end of the rail and focus with the convex lens alone: screen position P_1. Put the concave lens between the convex lens and P_1 and move the screen away until the image is sharp again: P_2.

Record: s = P_1 − lens position, di = P_2 − lens position.
Result: f = $(1/di - 1/s)^{-1}$, negative.

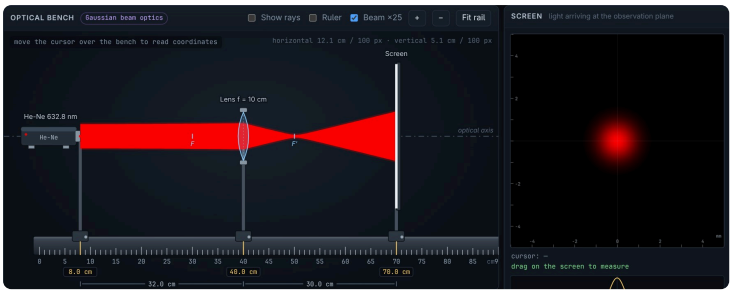


16 Concave Mirror — Focal Length

Slide the screen, in front of the mirror, until the reflected image is sharpest. Read the object–mirror and screen–mirror distances from the rail.

Record: do and di.
Result: 1/f = 1/do + 1/di, R = 2f.

LASER BEAMS



5 He-Ne Laser Through a Lens

Move the screen to find where the laser spot is smallest (the beam waist). “Beam ×25” draws the beam wider so that the focusing is visible; it changes nothing else.

Record: distance from the lens to the smallest spot.
Result: it lies very close to f.

DIFFRACTION & INTERFERENCE



6 Single-Slit Diffraction

Read the slit-to-screen distance from the rail. On the screen, drag across the central maximum, between the first dark bands on either side, to measure its full width.

Record: L and W.
Result: slit width a = 2λL / W.



7 Double-Slit Interference

Read the slit-to-screen distance from the rail. On the screen, measure across several bright fringes and divide by the number of spacings.

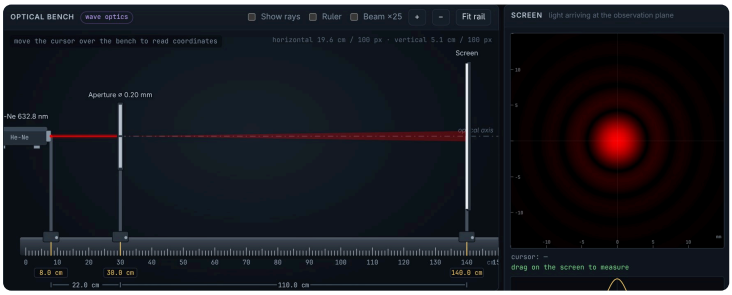
Record: L and Δy.
Result: λ = Δy · d / L.



11 Diffraction Grating

Read the grating-to-screen distance from the rail. On the screen, measure the distance from the central spot to the spot of order m.

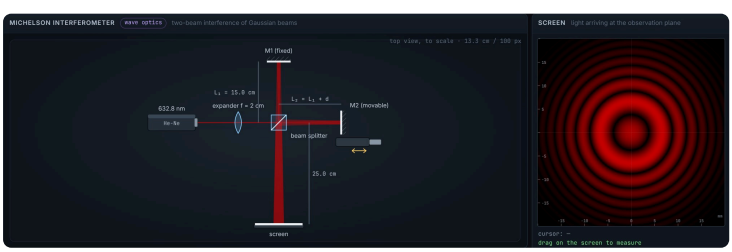
Record: L, y and m.
Result: λ = d sin θ / m, tan θ = y / L.



12 Circular Aperture — Airy Pattern

Raise the exposure until the first dark ring around the central disc is clear, then measure the diameter of that ring across the centre.

Record: L and D_ring.
Result: aperture D = 2.44 λL / D_ring.

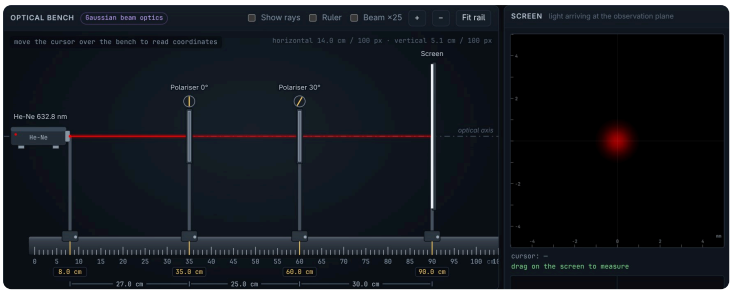


9 Michelson Interferometer

Reset the fringe counter. Move mirror M2 slowly with the fine control or “Scan” and count the fringes that pass the centre. Tilt M2 to turn the rings into straight fringes.

Record: mirror travel Δd and fringes N.
Result: λ = 2Δd / N.

POLARISATION

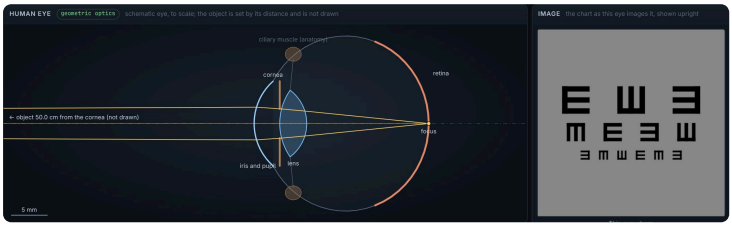


13 Polarisation — Malus's Law

Set both polarisers to the same angle and note the detector reading I_0 in the Physics panel. Turn the second polariser by θ and note the reading I. Go in steps of 10° up to 90°.

Record: θ, I and I_0.
Result: I/I_0 compared with $\cos^2\theta$.

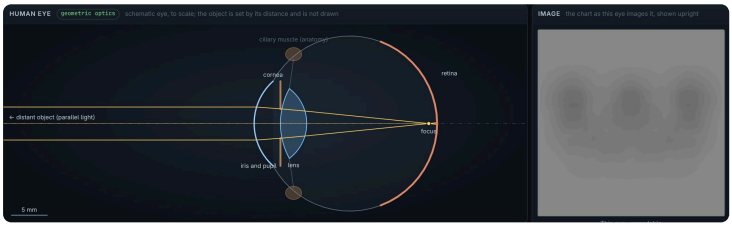
THE EYE



17 The Eye — Accommodation

Move the object closer and watch the lens thicken while the image stays sharp. Find the far point and the near point: the last distances at which the image is still sharp.

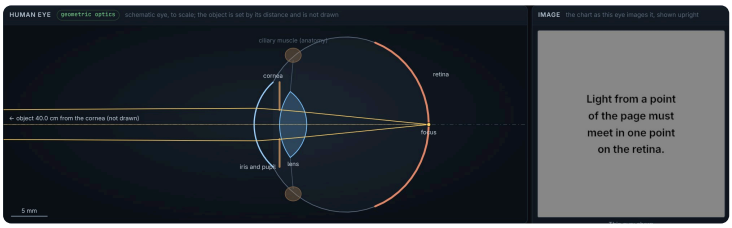
Record: near point and far point (0 if distant).
Result: A = 1/near − 1/far.



18 The Eye — Glasses

The eye is short-sighted and the distant chart is blurred. Put on glasses and change their power until the chart is as sharp as it gets. Repeat with "Hyperopia" and the lens held relaxed.

Record: glasses power Fs.
Result: refractive error K of the eye.



19 The Eye — Ageing

Read text at 40 cm and raise the age until it blurs. Find the near point at several ages. At 55, add positive glasses until the text is sharp again.

Record: age and near point.
Result: A = 1/near point; slope −0.30 D per year.

5. Exams in brief

Student. On the start screen choose *Join an exam*, enter your name and class, and join the classroom your teacher opened; both computers must be on the same Wi-Fi. Without a network, *Open an exam file* starts an exam from a file.

In a laboratory question the bench is in Experiment mode and the theory cannot be revealed: you answer from what you measure.

Teacher (teacher's edition). *Exams — teacher* is where you write an exam (written, multiple-choice and laboratory questions), open a classroom, start the exam for everyone, mark the answers and export the results.

The room set-up and the full procedure are in **docs/EXAMS.md**. The eye experiments have their own short guide, **docs/eye-guide.pdf**.

Screenshots taken from the application. The experiment steps, notebook entries and formulas are those the application itself shows in each experiment's Guide.